

Depth as sampling geometry, and the quorum as a 2×2 mechanism

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1 September 2026 — consolidates notebook §66–71 of `riemann-weil-phenomenology`

Abstract

Two results replace two earlier readings. (1) The depth law $-\ln \lambda_{\min} = s(\chi)\mu + b$ of the truncated Weil forms, previously described by a four-variable regression on fourteen Dirichlet characters that failed out of sample, is a *geometric* property of the zero set: the floor c_L of the form is the sampling constant of the zeros for the Paley–Wiener space of type $L/2$, made exponentially small by the desert $(-\gamma_1, \gamma_1)$ and by the gaps longer than the Nyquist step $2\pi/L$. A two-coefficient law fitted on ζ alone predicts $s(\chi)$ out of sample with median ratio 0.89 (10 of 14 characters within 20%), with no character-specific parameter. (2) The quorum theorem (every proper sub-Euler product violates positivity) has a mechanism: an unconditional 2×2 interlacing lemma, *depth* $\times$ *coupling*, whose three quantities we certify at $\mu = 11$ and measure at three scales; the coupling of the bottom vector to a missing prime decays in p at half the rate of its silence, which is what makes the determinant negative. Both are statements about explicitly defined forms; nothing bears on RH.

1 Depth is a sampling constant

Under RH, $Q_L(f) = \sum_{\gamma} \hat{f}(\gamma)^2$ with $\hat{f} \in PW_{L/2}$, so $c_L = \inf Q_L/\|f\|^2$ is a sampling constant of the zero set, positive by Beurling (note *sampling-floor*, Theorem 1). The finite-basis minima saturate: at $\mu = 3$, $\lambda_{\min}(N) = 1.026, 0.775, 0.641, 0.591, 0.571, 0.562 \times 10^{-7}$ for $N = 9 \dots 61$, converging geometrically to $c_{\log 3} \approx 5.55 \times 10^{-8}$; at $\mu = 11$, $3.59, 1.86, 1.54 \times 10^{-48}$ for $N = 47, 57, 67$.

Synthetic controls (zero-side Gram, $\mu = 3$). $Q_z = \sum_{\gamma} 2\hat{c}(\gamma)\hat{c}(\gamma)^{\top}$ reproduces the prime-side floor (ratio 0.93, the truncation tail). Three fictitious zeros placed in the desert raise $\lambda_{\min}$ by 9×10^6 ; a comb with the same γ_1 and density changes it by 0.23 only; shifting the whole spectrum changes $-\ln \lambda_{\min}$ linearly in the desert width with slope 1.45 against $L = 1.10$ for Slepian’s single hole. The desert is the cause; the arithmetic fine structure touches only the prefactor.

The geometric law. With $\nu = 2\pi/L$ the Nyquist step,

$$-\ln c_L \approx aL(\gamma_1 - \nu)_+ + bL \sum_k (\gamma_{k+1} - \gamma_k - \nu)_+, \quad a = 1.69, b = 0.82,$$

the two coefficients fitted on the μ -series of ζ only ($\mu = 3, 8, 11, 16$, within 8%; Slepian’s naive values would be 2, 1). Linearity in μ follows: the sub-Nyquist region extends to $\gamma^* \approx 2\pi\mu$ and the excess sum grows like μ .

Out of sample (coefficients frozen). Predicted s = slope of the law between $\mu = 11$ and 22, against the measured s of the depth note:

χ ratio	ζ 1.22	χ_3 0.82	χ_4 0.82	χ_5 0.87	χ_7 1.08	χ_8 1.00	χ_{11} 1.17	χ_{12} 0.88
χ ratio	χ_{13} 1.35	χ_{15} 1.02	χ_{19} 0.96	χ_{21} 0.65	χ_{24}^e 1.04	χ_{24}^o 0.38	χ_{-23} 0.89	

Median 0.89; ten of fourteen within 20%; no parameter per character. The downward bias is mechanical (zero caches end near $\gamma \approx 85 < \gamma^*(22) = 138$). The two failures have $\gamma_1 < \nu$: the linearized desert term switches off too abruptly, where Slepian's $1 - \lambda_0(c)$ does not. Where the four-variable map (trained on these characters) died out of sample (χ_{-23} : -29%), the geometric law predicts χ_{-23} to 0.89.

What the desert alone explains. Computing the Slepian constant to arbitrary precision: $1 - \lambda_0(\tau\gamma_1) = 3.35 \times 10^{-6}$, 5.42×10^{-14} , 2.93×10^{-16} at $\mu = 3, 11, 16$, i.e. 75%, 28%, 21% of the measured depth. The union of all sub-Nyquist gaps gives 74 of 110 nats at $\mu = 11$ but is not a bound in either direction. The sharp constant is c_L itself.

Conjecture 1. *The law holds with absolute a, b depending only on the window class, with the desert term replaced by $-\ln(1 - \lambda_0(\frac{L}{2}\gamma_1))$.*

2 The quorum as a mechanism

Write $Q_S = Q + T_M$, T_M the towers of the missing primes M .

Lemma 2 (unconditional). *Let v be a unit vector, $w = T_M v$, $\kappa = \|w - (v^\top w)v\|$, $u = (w - (v^\top w)v)/\kappa$, and $a = v^\top Qv + v^\top T_M v$, $d = u^\top Q_S u$. If $a d < \kappa^2$ then Q_S is indefinite.*

Proof. The 2×2 principal minor of Q_S on $\{v, u\}$ is $\begin{pmatrix} a & \kappa \\ \kappa & d \end{pmatrix}$ with negative determinant, hence a negative eigenvalue; Cauchy interlacing. $\square$

No eigenvector exactness and no hypothesis on zeros is used. Taking v the bottom vector of Q : $a \approx \varepsilon + \sum_{p \in M} \delta_p$ with $\varepsilon = \lambda_{\min}$ and $\delta_p = v^\top T_p v$ the *silence* of v to p ; κ is its *coupling*. Depth $\times$ coupling: this is the content of the lemmas B1/B2 of the note *lemma-quorum-scales*, in the form in which they are provable.

Certified instance, $\mu = 11$ (`code/quorum.2x2.py`, Q and T_p in Arb balls). $\varepsilon = 3.5832 \times 10^{-48} \pm 3 \times 10^{-53}$. $T_{11} \equiv 0$ ($\Theta(L) = 0$ at the window edge), so the voting primes are $\{2, 3, 5, 7\}$ — the $2^4 - 1$ of the certified theorem. All fifteen proper sub-products are certified indefinite by 2×2 certificates built on the *same* v ; worst margin $M = \{7\}$: $\kappa^2 = 3.8 \times 10^{-9}$ against $a d = 1.4 \times 10^{-16}$. This re-proves the $\mu = 11$ theorem independently of its 340 witnesses.

The μ -content. High-precision measurement at three scales, criterion checked on every proper subset:

μ	ε	δ_p ($p = 2, 3, 5, 7, \dots$)	κ_p	subsets	re
8	7.5×10^{-33}	0.052, 3.1e-4, 1.3e-10, 9.4e-21	0.70, 0.53, 3.6e-3, 8.8e-9	14/14	
11	3.6×10^{-48}	0.058, 4.8e-4, 1.2e-9, 7.4e-17	0.74, 0.77, 0.051, 6.2e-5	14/14	
16	1.9×10^{-73}	0.064, 6.6e-4, 4.9e-9, 4.0e-15, 2.0e-30, 1.2e-40	0.80, 0.88, 0.28, 6.7e-3, 1.5e-9, 2.5e-15	62/62	

Two laws in p at fixed μ : $-\ln \delta_p \approx 7p - 12$ and $-\ln \kappa_p \approx 4p$ ($p \geq 5$). *The coupling decays at half the rate of the silence*; $\kappa_p^2/\delta_p \geq 9$ everywhere and grows with μ (for $p = 7$: $8 \times 10^3 \rightarrow 5 \times 10^7 \rightarrow 10^{10}$). The worst case always contains 2 ($a \approx \delta_2 \approx 0.06$ against $\kappa^2 \approx 0.5$ –0.9); two preregistered guesses died (the silence grows mildly with μ ; the worst case is not the largest isolated prime).

What remains for every scale. Three unconditional analytic statements about the bottom vector: $\varepsilon(\mu) \leq e^{-s\mu+b}$ (the desert, Section 1), $\delta_p(\mu) \leq e^{-7p+c}$, $\kappa_p(\mu) \geq e^{-4p-c'}$. The first is the depth law; the other two are new, concern the autocorrelation of an explicit eigenvector at the points $\log p$, and are where an analyst can enter.

Status

Lemma proved; $\mu = 11$ instance certified; everything else measured with stated numbers. Neither section moves RH: the first describes how small the positive floor is, the second why removing a prime destroys it.